

The diagram illustrates a Docker Compose network architecture. It is divided into two main sections: **PUBLISHED PORTS** (Host-facing endpoints) and **DOCKER COMPOSE NETWORK** (Internal service discovery by DNS name).

PUBLISHED PORTS:

- :8088** API Gateway LB
- :3000** Grafana
- :9090** Prometheus
- :15672** RabbitMQ management

DOCKER COMPOSE NETWORK:

Internal service discovery by DNS name

The network components and their interactions are as follows:

- api-gateway-lb** (HAProxy) receives traffic from the host (published port 8088) and routes it to **api-gateway** (Spring WebFlux).
- api-gateway** routes **/auth** requests to **auth-service** (Spring Boot) and **/requests** to **request-service-lb** (HAProxy).
- request-service-lb** routes traffic to **request-service x 2** (Spring Boot).
- request-service x 2** connects to **postgres** (Two logical databases) and **redis** (DB 0 core · DB 1 gateway).
- outbox-publisher** (Publisher role) connects to **rabbitmq** (Broker).
- rabbitmq** connects to **request-worker x N** (Consumers).
- request-worker x N** connects to **prometheus + grafana** (postgres-exporter, redis-exporter, RabbitMQ metrics).

Ports map to the corresponding named containers.